Your Dreams Our Goal

POORNIMA UNIVERSITY

Member of Association of Indian Universities & Approved by UGC (Govt. of India) under 2(f) & 12(B)

Faculty of Engineering and Technology
Faculty of Computer Science and Engineering
Department of First Year
Synopsis on SIGN-AI

Topic of the project: SignAI :- An AI-Based Sign Language Detection System

Communication is a fundamental part of human interaction, but people with hearing or speech impairments often face difficulties in expressing themselves. Sign language is widely used by such individuals, but not everyone understands it.

This project aims to develop a system that can recognize hand gestures and convert them into readable text or speech using Artificial Intelligence. The system uses a camera to capture hand signs and processes them in real time.

Objective and Scope:

- To develop an intelligent system using Artificial Intelligence that can recognize sign language gestures in real time.
- To convert hand gestures into readable text and/or speech for easy understanding.
- To design a user-friendly interface that allows real-time interaction through camera input.
- To build a foundation for developing assistive technologies for differently abled people.

Scope:

- The project focuses on recognizing basic hand gestures or alphabets of sign language.
- The system can be implemented as a desktop or Android application using tools like Android Studio.
- The project can be expanded to support:
  - Full sentence recognition
  - Multiple sign languages
  - Voice output for better interaction

Process Description:

Start

↓

Camera Input (Hand Gesture)

↓

Hand Detection (OpenCV / MediaPipe)

↓

Preprocessing (Resize, Normalize)

↓

AI Model Prediction

↓

Recognized Sign

↓

Display Text / Voice Output

↓

End

Resources and Limitations:

Hardware Requirements:

- **Computer/Laptop with minimum 4GB–8GB RAM**
- **Webcam for capturing hand gestures**
- **Internet connection for running the web application**

Software Requirements

- **Programming Language: Python**
- **Web Technologies:**
  - **HTML, CSS, JavaScript (for frontend design)**
  - **Flask or Streamlit (for backend/web app development)**
- **Development Tools:**
  - **Jupyter Notebook or Google Colab (for model training)**
  - **Any code editor (VS Code, etc.)**
- **Libraries:**
  - **OpenCV (image processing)**
  - **MediaPipe (hand detection)**
  - **TensorFlow (model training)**
  - **NumPy (data processing)**

Limitations of the Proposed System:

- The system may have reduced accuracy if the dataset is limited or not welltrained.
- Real-time detection through a browser may be slower than native applications.
- The project mainly supports **basic gestures**, not complex sentences
- It may face issues in **low lighting or complex backgrounds**.

Conclusions:

The proposed system SignAI demonstrates an effective use of Artificial Intelligence, Machine Learning, and Computer Vision to address real-world communication challenges faced by hearing and speech-impaired individuals.

The project introduces an innovative approach by integrating real-time hand gesture recognition with a **web-based interface**, making the system easily accessible without the need for complex installations.

The main achievements of this project include:

- Successful detection and interpretation of hand gestures in real time
- Conversion of sign language into readable text (and optional speech output)
- Development of a user-friendly and accessible web-based system

An important feature that makes **SignAI** stand out is its **simplicity, realtime performance, and practical usability**, especially for educational and assistive purposes.

Name of Leader: Kunal Bairwa Name of Guide: Mrs. Priyanka Tiwari

Year/Semester: 1st Year/2nd Semester

Registration No.: 2025/22545

Details of Team Members: Prateek Sharma (23452)

Jhanvi kumar

Neha Singhal